

What a Complete Mechanism for a Λ -Tied MOND Law Must Do: A Closure Map of Scoped No-Gos and a Specification

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Abstract

The programme ties the MOND acceleration scale to the vacuum, $a_0 = \kappa c \sqrt{G \rho_\Lambda}$, with $\kappa = \frac{1}{2}$ fitted, never derived. Its best current effective law, candidate B, is the MOND kernel plus hierarchical ownership (only the outermost bound system carries the phantom) plus a cold component with the cosmic amount $\Omega_c h^2 \approx 0.12$: the mass is still required, and no dark-matter particle is added. No committed action produces B. This note is a map and a specification, not a result. (i) Three gaps separate B from a mechanism: a legal owned switch (Gap 1), the cold fluid that *is* the law’s dark density (Gap 2), and the a_0 – Λ tie with its coefficient (Gap 3). (ii) A Lean library of 331 theorems (standard axioms only) certifies, from stated premises, that ownership is not a function of local field data, that a continuous law with no external-field effect is linear, that a length $\propto M^{1/2}$ needs an acceleration scale, and that an AQUAL action gives the kernel in spherical symmetry; it certifies no empirical fact. (iii) We tabulate 25 committed routes to the missing object: 22 are scoped no-gos on their frozen classes, two owner-directed hand-checks find no mechanism, and one shows realisability only. None is a mechanism. (iv) The failures sharpen the target into a specification: an early cold fluid present before structure forms (a dust created at turnaround supplies $\Omega h^2 = 0.1200 \times 10^{-1376.8}$ at $z = 1100$); ownership that remembers boundness (a causal latch keeps $n = 0.791$ at one free-fall time against the required 0.99); a distribution that carries the $M^{1/2}$ scale itself (passive bound infall scales as $M^{0.33-0.34}$, and relaxation at a vacuum-set rate reaches $\Gamma \tau \leq 0.79$ against the ≈ 4.1 – 4.2 needed); a target that is field-local in the Newtonian field but needs bound-domain labels; and no created rest mass. (v) B’s ultra-faint failure (3.5 – 3.9σ) does not move under the 2026 binary-corrected and DEIMOS dispersions, and the bound-core (Λ CDM-comparator) reading is preferred at satellites by small margins: a lean, not a detection. (vi) $\kappa = \frac{1}{2}$ is fitted (0.465 ± 0.076 from the baryonic Tully–Fisher relation), and Milgrom’s $cH_0/2\pi$ footing fits SPARC as well. (vii) Gaia DR4 (2 December 2026) is a survival test of B against the bare MOND law; B, Λ CDM and Newton all predict $\hat{\gamma} = 1.000$.

1 The law, candidate B, and the three gaps

The law. $a_0 = \kappa c \sqrt{G \rho_\Lambda}$ with $\kappa = \frac{1}{2}$ fitted; two footings are carried, canonical $9.36\text{e-}11$ and alternative $1.13\text{e-}10 \text{ ms}^{-2}$ (written can/alt). With the monotone kernel ν_{mono} the law fits SPARC’s radial acceleration relation with an rms of 0.1003 dex, against 0.2755 for Newton (CFG4, re-run clean). The distinctive galaxy-scale content and the high-redshift side (a flat $a_0(z)$, which a decisive test would need baryon masses calibrated to 0.10 dex to see) are reported in PAPER38 [P38] and not repeated here.

Candidate B is an effective law: the kernel, a hierarchical ownership rule (a top-level bound system carries the phantom; an owned subsystem is Newtonian internally), and a conserved cold fluid with $\Omega_c h^2 = 0.1200$ whose amount is fitted. The cold mass is still required; it is not a particle species in any lane. The status page’s bottom line is that “no committed action produces candidate B” (`campaign_fresh_gravity/STANDING_2026-09-29.md`).

Gap 1, the legal owned switch. A gate multiplying a shift-symmetric field leaves the dynamical mass equal to the baryonic mass beyond its edge (Gauss); local gates are unstable in 44 of 48 cases; a nonlocal enclosed-mass gate is stable in 48 of 48 with no new constant, but it is bilocal (not a legal local term) and its edge sits at 0.11 – $0.24 r_{\text{ta}}$, below B’s window (CFG48); a dynamical gate scalar is stable only with two new untied constants (CFG49).

Gap 2, the cold fluid that is the law’s dark density. For a spherical baryon profile the fluid must satisfy $C(r) = \rho_c r^3 g_{\text{tot}} = (a_0/4\pi) M_b(< r)$ (CFG44). No local or equation-of-state closure gives it, the tidal-tensor closure fails (CFG50), and written as a memory-kernel or light-cone exchange it reacts on the baryons at 0.065 – $22.5 g_{\text{law}}$ and costs 23 – $73\times$ their orbital energy in one r_{ta} convention, 57 – $318\times$ in B’s committed one (CFG70, CFG72). A documentary count of 54 committed constructions finds none that derives the dispersion from an action (D = 0, P = 11, I = 5, X = 27, U = 11). In this record Gap 2 reduces to Gap 1’s object, as an identification, not a theorem.

Gap 3, the tie. The tie can act on a conserved fluid’s stress cap with no new local degree of freedom, but the cap’s entry is postulated (CFG43), and a barotropic saturating cap engages over a window of $11\times$ up to $1\text{e}4$ – $1\text{e}5$ in mass, depending on convention. The coefficient κ is the subject of Section 6.

2 What Lean certifies

The ChainCert library (fable_independent_2026/lean_2026/ChainCert/) builds with its verifier reporting 331 theorems checked, 0 with non-standard axioms and 0 sorry mentions; MUTATE=1 fails as required. The modules used here were added in three steps: Ownership, Theory, AONumeric and DoorEleven (47 theorems, 219 in the library), Action, Dimension and FluidLink (62 theorems, 281), and CalibrationWall (50 theorems, 331). *Lean certifies implications from premises, not data.* $\kappa = \frac{1}{2}$, the kernel, ρ_Λ constant, the hierarchy and the ownership rule enter as named premises.

- **Ownership is non-local.** If $\nu \neq 1$, a top-level and an owned system with the same internal and external fields get different internal accelerations, so no function of the field data reproduces the rule (ownership_distinguishes, ownership_not_field_local); B's own kernel meets the premise (CandidateB.ownership_nonlocal).
- **No external-field effect forces linearity.** A continuous map with $F(z + z_e) - F(z_e) = F(z)$ is linear (noEFE_continuousLinear); hence no pointwise law with MOND's deep limit is EFE-free, in scalar or vector form (law_not_efeFreeRay, vecLaw_not_efeFree). This covers pointwise (QUMOND-type) laws only, not the AQUAL or QUMOND field equations. Ownership escapes only because it is non-local.
- **The dimension theorem.** No monomial in (G, M, c) is a length $\propto M^{1/2}$; with one extra constant X such a length exists if and only if (G, c, X) form an acceleration scale. What is forced is an acceleration scale, not that X is an acceleration ($\hbar$ is admissible). Monomials only.
- **The AQUAL action gives the kernel.** For an AQUAL μ with $x\mu(x)$ increasing onto $(0, \infty)$, the spherical Gauss form $\mu(g/a_0)g = g_N$ is equivalent to $g = \nu(g_N/a_0)g_N$, and P2's kernel follows from $\mu = (\sqrt{1 + 4x^2} - 1)/(2x)$. Premises: the reduced Euler–Lagrange equation and regularity; spherical symmetry only. Ownership and the cold-mass rule stay postulates.
- **The calibration wall (CFG240).** In the deep limit g_{obs} depends only on $f \cdot a_0$ (f a baryon calibration); for exact P2 two distinct g_{bar} identify (f, a_0) ; with f free, $\sigma(\log a_0) \geq 3\sigma/\sqrt{N}$ (conditional on local slopes in $[0, \frac{1}{2}]$). These are the theorems behind PAPER38 [P38].

3 The routes tried, and why each fails

Each route below was given frozen criteria committed before its scripts, and is scored against the shared gates of closure_map/TEN_DOORS_GATES_2026-09-29.md: G1 reproduces $C(r)$ within 10% for $x = r/r_M \in [0.1, 30]$ over 10^9 – $10^{12} M_\odot$ with the same constants; G2 cold linear growth and the CMB; G3 reaction $\leq 0.10 g_{\text{law}}$ and energy $\leq$ the orbital energy; G4 no constant beyond κ and $\Omega_c h^2$; G5 well-posedness and the Solar System. Every FAIL is a scoped no-go on the frozen class only. Counting the rows honestly gives 25 routes: 22 scoped no-gos, two owner-directed phase-1 hand-checks that find no mechanism (11D and CFG253), and one realisability result (door 5). Owner-directed lanes (marked [†]: every door-11 variant including 11D, the four swings CFG242–CFG245 of 2026-10-01, and CFG253) are credited as such, following closure_map/RESEARCH_DIRECTION_2026-09-29.md; the AI sessions designed, froze, ran and reported them, failures included. Before the doors, the Gap 1 and Gap 2 constructions of Section 1 (CFG44, 48, 49, 50, 70, 72) had already failed; they are not counted again.

route	lane, commit	binding failure	class
1 Mashhoon-type kernel	CFG120, 9e4757627	a fixed-length linear response gives $R \propto M_b$: a sqrt(1000) mass spread	no-go (G1)
2 Verlinde [V17]	CFG117, bb504274c	point mass $C_V/C_{\text{target}} = (1+x)/x$; Solar System misses by seven orders [HFB17]	no-go (G1, G5)
3 dipolar medium	CFG121, 16fca9acd	medium budget needs $Q^2/\kappa_I \gtrsim 140$ against $\leq 0.16 / 1.2$ from growth	no-go
4 superfluid	CFG122, b7d41c302	both branches miss the target; the MOND branch has $c_s^2 < 0$	no-go
5 collisional $f(E, L)$	CFG130, c1d719fbf	$f \geq 0$ exists but is fixed by the target: it encodes $C(r)$, not derives it	realisable only
6 secondary infall	CFG118, d0baef700	radial scale follows the turnaround ($M^{0.33}$), not $r_M \propto M^{1/2}$; C/C_{target} 4–360 at $x \approx 0.1$ (spheres), 0.05–0.21 at $x \approx 28$	no-go (G1)
7 fuzzy soliton	CFG119, 435cb43e9	cores flat against the target's $1/r$; growth needs $m \geq 1.28 \times 10^{-20} \text{ eV}$	no-go

route	lane, commit	binding failure	class
8 interacting vacuum	CFG131, aa0d95aef	the forced equation of state differs $32\text{--}353\times$ across $10^9\text{--}10^{12} M_\odot$; $Q = 0$ in a stationary halo	no-go
9 Deser–Woodard / RR	CFG123, 9e4757627	the localised response is a negative density about 10 orders too small	no-go
10 mimetic	CFG124, 9e4757627	no stationary state; a ghost for $0 < gt < 2/3$	no-go
11A inflow [†]	CFG171, 1c91164e9	the only working flow law is AQUAL restated; the literal medium costs $16\text{--}205\times$ the orbital energy	no-go (restatement)
11B' directional [†]	CFG173, e415659c7	a $w = -1$ medium cannot flow; a massless flow's own gravity is $\leq 1.2 \times 10^{-6}$ of the phantom; a push is Le Sage gravity	no-go
11C-a timelike flow [†]	CFG172, f1585212e	the kernel is declared, not derived; Solar-System quadrupole tail $6.3 \times 10^3\times$ the bound	no-go (restatement)
11C-b single $F(K)$ [†]	CFG172, f1585212e	G1 needs $t \lesssim 10^{-6}$, G6/G7 need $t \gtrsim 4 \times 10^6$	no-go
11C-c matter- θ [†]	CFG172, f1585212e	the coupling supplies no law, and $c_s^2 < 0$	no-go
11C-d V0 region gate [†]	CFG172D, 249fa4ec8	blind in 96/96 cells or unstable: c_{eff} 4e3–9e3 km/s against a 37 km/s line	no-go
11D one-time pass [†]	CFG251, eff33f041	set-at-turnaround reproduces the ownership classes, not the amount: a literal freeze is $15.6\times$ short at the edge	no mechanism (hand-check)
12 covariant emergent gravity	CFG231, bb7d14eb8	G1 fails for all eight variants; the one that restates P2 has a Solar tail $Q_2 = 6.1 \times 10^3\times$ the bound and a ghost sign	no-go
13 BIMOND [M09]	CFG232, 982fe61dd	G1 passes only by inverting the target (M1-inv); a cold component in g fails, crossing 1.10 at $x = 0.483$	no-go
ownership latch (route A) [†]	CFG242, e04b22b5a	a static bound system has no memory: $n(t_{\text{ff}}) = 0.791$, minimum 0.143, against the required 0.99	no-go (G0)
foliated BIMOND (route B) [†]	CFG242, e04b22b5a	not strongly hyperbolic ($c_T^2 = 0$: $\partial_t^2 h = 0$) and undetermined (nullity 2)	no-go (G5)
turnaround-created dust [†]	CFG243, e8b702457	at $z = 1100$ the class supplies $\Omega h^2 = 0.1200 \times 10^{-1376.8}$; $2.8\text{--}38\times$ short even at $z = 10$	no-go (COSMIC)
bound early fluid [†]	CFG244, 50c8c7267	the bound mass scales as M^p , $p = 0.33\text{--}0.34$, not $\frac{1}{2}$; R_s/r_M spreads $3.0\text{--}3.2\times$ against 1.22 allowed	no-go (Gate A)
vacuum-rate relaxation [†]	CFG245, 48ef9f534	$\Gamma\tau = 0.79 (H_\Lambda)$, $0.14\text{--}0.16 (a_0/c)$, $0.27\text{--}0.33 (\sqrt{G\rho_\Lambda})$ against $\approx 4.1\text{--}4.2$ needed	no-go (Gate T)
dark energy into cold mass [†]	CFG253, c89139c1b	continuous transfer makes at most about 3% of the cold mass; an early transfer is CDM with a vacuum origin story	no mechanism (hand-check)

What the referees reproduced. The ten doors' headlines were re-derived by independent referee lanes (CFG151–CFG159) with their own code; none changed a verdict, but “referees were not blind to the README numbers”, several cells rest on sub-lines the referees did not test, and door 2 has no referee. CFG188 re-derived the field-equation core of 11C-a, b and c; 11C-d has no independent re-derivation. CFG242–CFG245 were re-run in place by a second session from the committed state, which checks the code path, not the model. CFG244's Gate A imports CFG118's code and reproduces it for the bound subset: it is *not independent*. CFG242's frozen triage scored 20 untested routes at $P(\text{all gates}) \leq 1.3\text{e-}3$ each before choosing two.

Two owner-directed readings of the vacuum side are not routes to the missing object and are not counted: dark energy as the void–wall expansion difference (CFG252, NON-DIAGNOSTIC; it loses the flat $a_0(z)$) and an outside contact with the dark energy (CFG254; “nothing before recombination” is contradicted by BBN, the acoustic peaks and N_{eff}).

A cross-door pattern, a reading and not a theorem. Where cold matter is dynamical, its scale comes from the turnaround, a particle mass or a free time scale, never $r_M \propto M^{1/2}$ (doors 6, 8, 10, and CFG244, CFG245). The day's four swings (CFG242–245) add that ownership must remember boundness and that the fluid must exist before structure.

4 The specification

Each item is a necessary condition read off the failures of scored classes; none is shown sufficient, and the set is not shown jointly satisfiable. CFG230 lists twelve requirements of the same kind (R01–R12) and finds that only R01 and R02 are met by any scored row as a mechanism.

- S1. **An early cold fluid, present before structure.** The fluid must exist with the cosmic amount at recombination. A conserved dust created at turnaround supplies $\Omega h^2 = 0.1200 \times 10^{-1376.8}$ at $z = 1100$, and in a baryon-only universe the trigger never fires (CFG243; the baryon-only σ_b is 0.076 against the threshold 1.276). A turnaround source can at most add to or relabel the early fluid. Continuous vacuum-to-fluid transfer after recombination can make only a few per cent of it (CFG253). The premise of the conservation rule [P36] is exactly what such classes cannot supply.
- S2. **Ownership that remembers boundness, on that early fluid.** A causal doubled-field latch passes causality and bound-only rows but forgets a static bound system ($n(t_{\text{ff}}) = 0.791$ against 0.99); a post-hoc energy-driven repair needs 5.0 free-fall times and an untied coupling (CFG242). A first-crossing rule gives ownership to the smallest progenitors: only 0.16–0.21 of the baryons first turn around in a system of at least 0.1 of the final mass (CFG243, post hoc). Ownership is not field-local (Lean, Section 2).
- S3. **A distribution that carries the $M^{1/2}$ acceleration scale itself.** By the dimension theorem an acceleration scale must enter. Passive infall does not supply it ($M^{0.33}$; CFG118, CFG244), and a relaxation at a vacuum-set rate is too slow: the 10% line needs $\Gamma \geq 5.37 / 5.17 H_\Lambda$ (can/alt) at $\tau = t_0$, and partial relaxation misses the RAR by 0.15–0.28 dex against the 0.10 line (CFG245). The galaxy and cluster windows nearly meet near $1.7\text{--}1.9 H_\Lambda$, a rate no vacuum candidate provides (arithmetic, not a finding).
- S4. **Field-local in the Newtonian field, but labelled by bound domains.** The law’s local-field (QUMOND phantom) density is a function of the Newtonian field, but it is not the CFG44 target for extended baryons (ratio up to 2.33 for P2), and the local form does not give ownership (internal-to-Newtonian ratio up to 3.16 where ownership needs 1): the local target needs non-local labels, a bound domain and a supply cap (CFG245, G0).
- S5. **No created rest mass.** The heat ledger is not the obstacle (at most $1.9\text{e-}4$ of the vacuum energy of the ball, CFG242), but for any class that creates rest mass the ledger is: $f(30) = 24.8$ in one r_{ta} convention, and 1.7 against 1 on the Lagrangian-volume ledger (CFG243, CFG131).
- S6. **The inherited gates.** Cold at $z \gtrsim 10$ to $k = 30/\text{Mpc}$ ($c_s^2 \leq 4.608\text{e-}12$, CFG230 R05); reaction and energy within G3 (exchange forms give $22.5 g_{\text{law}}$, CFG242); no new constant (G4); an external-field branch chosen explicitly (ownership, or an EFE that DR4 can see); and a Solar tail handled by ownership or by screening $\geq 3.5\text{e}3$ for any kernel with a tail $\geq 0.28205 a_0$ (CFG230 R09).

In one sentence: the target is an early cold fluid with the cosmic amount, owned by a memory-carrying, bound-domain assignment, whose distribution carries the $M^{1/2}$ scale itself; neither passive infall nor vacuum-rate relaxation supplies it, and no committed construction derives it.

5 Satellites: where B already fails

The failure. Under ownership an accreted satellite obeys the isolated law of its infall baryons. For the Milky Way ultra-faints that law under-predicts the dispersions: the Kaplan–Meier median offset is $+0.3245 \pm 0.038$ (3.77σ , canonical) with the nine upper limits restored and a 0.077-dex systematic floor (CFG28); tides and noise do not explain it, and the published binary bounds do not either (CFG29). The ΛCDM side is weakly diagnostic here: its pass window spans 3.5–5 decades of halo mass.

The 2026 dispersions (CFG259). Re-scoring with the DEIMOS archive [G26] and the binary corrections of [AP26] moves B’s median by at most 0.005 dex; the significance stays 3.5 to 3.9σ on both footings. The corrections reach 8 of 40 systems, with a median of -0.068 dex (free binary fraction) or -0.104 dex ($f = 0.7$), a fifth to a third of the offset. The one place they bite is CFG29’s weak-binary subsample, which drops to $2.8\sigma / 2.6\sigma$ under the $f = 0.7$ correction. The Boötes I conflict between the multi-epoch value and the single-epoch correction changes none of this.

Which reading the satellites prefer (CFG244 Gate H, re-scored by CFG259). Reading (a) is “satellites own nothing” (B’s isolated law); reading (b) is a bound cold core, which is the record’s ΛCDM comparator (a stellar-to-halo relation [M13] with NFW concentrations [D08]). On the committed values (b) is preferred by $\Delta\chi^2 +13.58 / +11.32$ (P2 kernel, record’s error model), but the frozen class is H4 (ambiguous) because the record’s own kernel falls to $+8.60$ in the strict view on the alternative footing. Under every 2026 variant both kernels cross the line and the class moves to H2; the margins are small (the variant DV3 sits only 0.24 above 9), the planted-sample power is 0.28, the whole preference comes from the ultra-faints (without them $\Delta\chi^2$ is about -1.3), and 33 of the 40 ultra-faints take the clamped stellar-to-halo value $10^9 M_\odot$, an extrapolation.

What this implies for the ownership rule. If the lean held, an accreted satellite would keep a bound cold core

of order $10^9 M_\odot$ rather than obey the isolated law of its stars: the rule would have to let satellites own a bound remnant of the early fluid. The class cannot supply that collapse mass from its own constants, and at hosts the same class fails Gate A; H2 says that the bound-core route and B differ at satellites, “not a lean for the framework”. It says nothing about hosts, and it is a lean, not a detection.

6 Gap 3: the coefficient

κ is fitted. The measured values are 0.465 ± 0.076 from the baryonic Tully–Fisher relation and 0.547 ± 0.175 distance-free; $\frac{1}{2}$ lies inside both, and the Unruh/de Sitter route is retired ($n = 2$ gives $\kappa = 1.447$, 12.9σ off). On the footing question, Milgrom’s $cH_0/2\pi$ fits SPARC as well as the framework ($\Delta\chi^2$ 5.34 against 7.04 alt), and “beats $1/2\pi$ ” holds only on the ρ_Λ footing. In the $\kappa c\sqrt{G\rho}$ convention Milgrom’s value is $\kappa = \sqrt{2/(3\pi)}$ (Lean, Footing).

The 32π puzzle. Writing $C = \Lambda_{\text{geom}}/a_0^2 = 32\pi$ into the canonical-clock construction forces $R = G_{\text{cosm}}/G_N < 0.824$, while BBN alone gives $G_{\text{BBN}}/G_0 = 0.98 \pm 0.06$ [A20]: that branch is excluded, conditionally on its transfer assumptions, and $C > 1198$ follows at $R \geq 0.92$ (a lane result the orchestrator did not re-run). This is a conditional no-go for one route to 32π , not for 32π itself; the later variable-clock results are toy-level, and the rational 4 in $G\rho_\Lambda = 4a_0^2$ stays underived. Another agent’s note (`gemini_pi_puzzle/SOLVING_THE_PUZZLE.md`), which says the puzzle is solved, is read on the status page as an interpretation, not a derivation. It sets $R^* \equiv c/\sqrt{G\rho_\Lambda}$ with $R^{*2}\Lambda = 8\pi$ and $a_0 = c^2/(2R^*)$; the algebra is correct, but the step that borrows a horizon surface gravity at a radius that is not a horizon is postulated. The status page’s wording stands: a coherent physical reading of the 32π , not a derivation of $\kappa = \frac{1}{2}$.

7 What would decide the rest

Gaia DR4 wide binaries, 2 December 2026. B predicts $\hat{\gamma} = 1.000$; the bare law’s floor is 1.161 (alt 1.192). $\hat{\gamma} \geq 1.084$ kills B; $\hat{\gamma} \leq 1.077$ kills the bare law. With the frozen $\sigma_{\text{sys}} = 0.02$ the cap is 8.1σ , and 3σ needs about 4,300 pairs out of $\sim 30,000$ expected. A merged P2 law gives 1.063–1.127, which separates from ownership by 2.3–3.7 σ at $N = 30,000$. B, Λ CDM and Newton all predict 1.000, so for B this is a survival test, never a confirmation.

$a_0(z)$. The flat law against $a_0 \propto H(z)$ is limited by the baryon calibration, not by sample size [P38]. CFG245’s class would predict a_0^{eff} at $z = 2.5$ lower by 1.2–1.4 dex; that is a prediction of a failed class, not a data statement.

Satellites. The ultra-faint test is capped near 2.5σ by systematic floors; what would move it is multi-epoch, binary-tested dispersions for more systems [O26, MC10] and a collapse-mass floor that data can shrink. Gate H’s lean would need its power raised well above 0.28 before it could be called a preference.

8 Limitations

- **The menu was written knowing the target.** The doors, door 11’s variants and the four swings are not a blind sample of theory space; several frozen hand estimates were wrong and are kept in the lane READMEs (for example CFG243’s survival ratio, wrong in direction).
- **Referees were not blind.** The independent re-derivations saw the README numbers; CFG242–245 were re-run, not re-derived; CFG244’s Gate A is not independent of CFG118.
- **Scope.** Every failure is a scoped no-go on its frozen class. Untested across the doors: non-spherical baryons, mergers, relativistic completions, Boltzmann-code CMB runs, a growing baryon core, a non-Lorentz-invariant vacuum, and Hossenfelder’s covariant Lagrangian. Several gates fail post hoc, after a stop rule, and those cells are labelled post hoc in their lanes.
- **Declared control failures are kept,** for example CFG244’s energy-drift control and CFG245’s MG2 premise; they do not change the binding failures but are part of the record.
- **Literature.** Some literature facts in the lanes are from memory and unverified, as each lane states.
- **What is not claimed.** Nothing here is claimed closed; $\kappa = \frac{1}{2}$ is fitted; nothing says the data favour the framework or Λ CDM; the cold mass is still required, and no dark-matter particle is introduced. A lean is not a detection.

9 Reproducibility

Every number carried by an audit row of `PAPER39_audit.py` is re-read from a committed file at git HEAD: the script checks each value against its committed source (a regex with one capture group over `git show HEAD:<path>`) and checks that the value is printed in the tex block carrying its audit tag. The tex side is checked at block level, not sentence level. It has two mutation modes (an altered expected value; an altered tex literal) that must fail. Numbers quoted only in running text and the references' journal data are not checked by the script. The sources it reads are the status page, the closure-map files, the lane READMEs of CFG28, CFG29, CFG118, CFG131, CFG230, CFG231, CFG232, CFG242–CFG245, CFG251, CFG253 and CFG259, the ChainCert README and verifier output, and the two notes cited for κ . No new analysis is done here.

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